Topology: MAT307/M405/MAT803
Problem Set 2 — Solutions to Problems 9–11

Conventions. A subspace $Y \subseteq X$ carries the topology $\{Y \cap U : U \text{ open in } X\}$. A *neighborhood* of x is an open set containing x. The closure $\overline{A}$ is the intersection of all closed sets containing A; it is the smallest closed set containing A.

Problem 9

Claim. If V is closed in Y and Y is closed in X, then V is closed in X.

Proof. Since V is closed in the subspace Y, the set $Y \setminus V$ is open in Y. By definition of the subspace topology, there is an open set $U \subseteq X$ with

$$Y \setminus V = Y \cap U.$$

Taking complements inside Y gives

$$V = Y \setminus (Y \setminus V) = Y \setminus (Y \cap U) = Y \cap (X \setminus U).$$

Here $X \setminus U$ is closed in X because U is open in X, and Y is closed in X by hypothesis. Hence V is an intersection of two closed subsets of X, so V is closed in X. □

Problem 10

Claim. Let Y be a subspace of X and $A \subseteq Y$. Then the topology τ_1 that A inherits as a subspace of Y equals the topology τ_2 that A inherits as a subspace of X.

Proof. Explicitly,

$$\tau_1 = \{A \cap W : W \text{ open in } Y\}, \qquad \tau_2 = \{A \cap U : U \text{ open in } X\}.$$

$\tau_1 \subseteq \tau_2$: Let $A \cap W \in \tau_1$ with W open in Y. Then $W = Y \cap U$ for some U open in X. Since $A \subseteq Y$,

$$A \cap W = A \cap (Y \cap U) = (A \cap Y) \cap U = A \cap U \in \tau_2.$$

$\tau_2 \subseteq \tau_1$: Let $A \cap U \in \tau_2$ with U open in X. Then $W := Y \cap U$ is open in Y, and since $A \subseteq Y$,

$$A \cap U = (A \cap Y) \cap U = A \cap (Y \cap U) = A \cap W \in \tau_1.$$

Therefore $\tau_1 = \tau_2$. □

Problem 11

Claim. Let $A \subseteq X$ and $x \in X$, and let $\mathcal{B}$ be a basis for the topology of X. The following are equivalent:

(i) $x \in \overline{A}$.

(ii) Every neighborhood of x intersects A.

(iii) Every basis element containing x intersects A.

Proof. **(i) $\Rightarrow$ (ii).** We prove the contrapositive. Suppose some neighborhood U of x satisfies $U \cap A = \varnothing$. Then $A \subseteq X \setminus U$, and $X \setminus U$ is closed since U is open. As $\overline{A}$ is the smallest closed set containing A, we get $\overline{A} \subseteq X \setminus U$. Since $x \in U$, $x \notin X \setminus U$, hence $x \notin \overline{A}$.

(ii) $\Rightarrow$ (iii). Every basis element $B \in \mathcal{B}$ is open, so a basis element containing x is a neighborhood of x. By (ii) it intersects A.

(iii) $\Rightarrow$ (i). Again by contrapositive. Suppose $x \notin \overline{A}$. Then $U := X \setminus \overline{A}$ is open (since $\overline{A}$ is closed) and contains x. Because $\mathcal{B}$ is a basis, there is $B \in \mathcal{B}$ with $x \in B \subseteq U$. Then

$$B \cap A \subseteq U \cap A \subseteq (X \setminus \overline{A}) \cap A = \varnothing,$$

since $A \subseteq \overline{A}$. So B is a basis element containing x that does not intersect A, and (iii) fails.

Thus (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i), and the three statements are equivalent. $\square$